Sohail YouTube Academy

Learn. Create. Grow.

Project Guide

Course bechne wali website — kya banaya gaya hai, kaise kaam karta hai, kaun si technology lagi hai, aur step-by-step kaise setup hua. Yeh document developer aur non-technical dono ke liye likha gaya hai.

PROJECT
Sohail YouTube Academy

TYPE
Course-selling website (single course)

STACK
Next.js 15 · Node.js · PostgreSQL

HOSTING
Vercel + Neon Postgres

REPOSITORY
`github.com/preeti28dec/Youtube-Academy`

DOCUMENT DATE
13 October 2026

SECTION 1

Project kya hai

Yeh ek **course bechne wali website** hai. Sohail apna YouTube course online bechte hain — student website par aata hai, payment karta hai, aur turant course dekhna shuru kar deta hai. Beech mein kisi ko kuch manually karne ki zaroorat nahi padti.

Poora flow — ek line mein

Student site par aata hai › Payment karta hai › Account apne aap banta hai › Course unlock › Videos dekhta hai › Progress save hoti hai

Sabse khaas baat

Videos secure hain

Jisne paise nahi diye, woh video nahi dekh sakta — chahe usko link hi kyun na mil jaaye. Yeh is project ki sabse zaroori requirement thi, aur poora system isi ke aas-paas bana hai. Detail Section 6 mein hai.

Design kahan se aaya

Design ek prototype se liya gaya hai — wahi colors, wahi spacing, wahi text. Rang indigo (`#4f56e3`), font Inter, look saaf aur bharosemand — taaki site tez chale aur maintenance sasta rahe.

SECTION 2

Kya-kya bana hai

Client ne 12 features maange the. Saare bane hue hain:

#	FEATURE	KAISE KAAM KARTA HAI
01	Home page	Course ka parichay, benefits, price, Enroll Now button, curriculum, mentor, FAQ
02	Course details	Poora description, "aap kya seekhenge", aur saare modules/lessons duration ke saath
03	Payment page	Naam, mobile, email + UPI / Card / Net Banking
04	Auto account	Payment confirm hote hi account banta hai, course khulta hai, password email par
05	Student login	Email *ya* mobile + password
06	Dashboard	Kharida hua course, saare modules, aur "Continue learning"
07	Video course	Videos website ke andar hi, secure player mein
08	Progress tracking	Har lesson par "Mark as complete", dashboard par percentage
09	WhatsApp support	Har page par floating button, pehle se likhe message ke saath
10	Admin panel	Module/lesson add-edit-delete, video jodna, students ki list
11	Mobile friendly	Saare pages mobile ke liye pehle design kiye gaye
12	Simple design	Koi bhaari animation nahi — site tez, maintenance sasta

SECTION 3

Technology — kya use hua aur kyun

CHEEZ	KYA CHUNA	KYUN
Frontend	Next.js 15 + React 19	Tez, SEO-friendly, aur Google par achhe se dikhta hai
Backend	Node.js (Next.js ke andar hi)	Frontend aur backend ek hi app mein — ek deploy, ek server, ek bill
Database	PostgreSQL (Prisma ORM)	Paisa involved hai, isliye transactions aur constraints zaroori
Styling	Tailwind CSS v4	Design tokens ek jagah, CSS file chhoti rehti hai
Login	JWT + httpOnly cookie	Token JavaScript se padha nahi ja sakta — XSS se surakshit
Password	bcrypt	Password kabhi seedha save nahi hota
Payment	Razorpay (Cashfree bhi chalega)	UPI, Card, Net Banking — sab ek jagah
Hosting	Vercel	Git push karte hi deploy, free SSL, auto scaling
Database hosting	Neon (Vercel ke andar se)	Free tier, Singapore region — India ke paas

Database kyun PostgreSQL

Is site ka saara data aapas mein juda hua hai:



1. **Paisa involved hai.** "Payment hua → account bana → access mila" — teeno ek saath hone chahiye, ya ek bhi nahi. Postgres transactions yeh guarantee dete hain.
2. **Duplicate rokna zaroori hai.** Database khud double enrollment reject karta hai, chahe payment ka confirmation do baar aa jaaye.
3. **Reporting aasaan.** "Is mahine kitne students" — ek query. Admin panel ke stats isi par chalte hain.
4. **Sasta aur aam.** Har hosting provider deta hai, free tier mil jaata hai.

MySQL bhi chal jaata. **MongoDB is project ke liye theek nahi** — data relational hai, aur payments mein constraints ka database-level par hona zyada safe hai.

SECTION 4

Database mein kya-kya rakha hai

TABLE	KYA RAKHTA HAI
User	Student aur admin dono. Naam, email, mobile, password (hashed), role
Course	Course ka naam, description, price (paise mein — decimal ki galti se bachne ke liye)
Module	Course ke chapter, sahi order mein
Lesson	Har video — title, duration, free preview hai ya nahi, video ka reference
Order	Payment ka record — kitna, kis method se, gateway ki id, status
Enrollment	**Access ka single source of truth.** Yeh row hai to course khulta hai, warna nahi
Progress	Kis student ne kaun sa lesson complete kiya

Do zaroori rules database mein hi likhe hain

- Ek student ek course mein **sirf ek baar** enroll ho sakta hai
- Ek lesson ka progress **sirf ek baar** save hota hai

Yeh rules application code mein nahi, **database mein** hain. Iska matlab: code mein koi bug bhi aa jaaye, to bhi duplicate data nahi ban sakta.

SECTION 5

Payment kaise kaam karta hai

Student details bharta hai › Gateway par payment › Gateway ka webhook › Account + access › Login details email par

Sabse zaroori baat

Course ka access **browser ke "payment successful" message par nahi** diya jaata. Woh message koi bhi nakli bana sakta hai. Access tab milta hai jab **payment gateway ka server khud** hamare server ko confirm karta hai (webhook), aur uska digital signature verify ho jaata hai.

Demo mode

Razorpay ki keys na hon to app "demo mode" mein chalta hai — poora flow waisa hi chalta hai (form, order, account banna, course khulna), bas asli paisa nahi katta. Keys daalte hi real ho jaata hai; code badalne ki zaroorat nahi.

SECTION 6

Videos kaise secure hain

Client ka sabse bada sawaal yahi tha: *"kya koi doosra vyakti direct link se course videos dekh sakta hai?"* Jawab: **nahi**. Teen layer lagi hui hain.

1. **Video ka link page mein hota hi nahi**
   Page ka source code dekhne par video ki file ka pata nahi milta. Player har baar server se naya link maangta hai.

2. **Server pehle check karta hai ki payment hua hai**
   Enrollment record hi access ka proof hai. Free preview lessons sabko milte hain, baaki sirf enrolled student ko.

3. **Link sirf 2 minute chalta hai, aur sirf usi student ke liye**
   Link ke andar lesson, student aur expiry — teeno cryptographically signed hote hain. Link copy karke kisi aur ko bhejne par server saaf mana kar deta hai.

4. **Har student ka watermark**
   Video ke upar student ka email halka sa dikhta rehta hai. Koi recording leak kare to pata chal jaata hai kisne ki.

Yeh test kiya gaya hai

TEST	RESULT
Apne account se video chalana	Chalti hai
Wahi link kisi aur ko bhejna	403 — nahi chalti
Wahi link dusre logged-in user se	403 — nahi chalti
Purana (expire) link	403 — nahi chalti
Link ko thoda badal kar try karna	403 — nahi chalti
Video file ka URL seedha kholna	404 — milti hi nahi
Video beech se aage khinchna (seek)	Theek chalta hai

Imaandari se ek baat

Duniya ka **koi bhi** system mobile se screen recording ko 100% nahi rok sakta — Netflix aur Udemy bhi nahi. Lekin DRM kai devices par recording ko black screen kar deta hai, aur watermark se leak karne wala pehchana ja sakta hai. Isliye leak ka risk practically bahut kam ho jaata hai. Jo bhi "100% impossible" ka vaada kare, woh sach nahi bol raha.

SECTION 7

Setup kaise hua — step by step

Project zero se yahan tak kin kadmon se pahuncha:

Part A — Code

1. **Design se code**
   Prototype se saare colors, spacing aur text nikaal kar Tailwind design tokens banaye, taaki poori site ek jaisi dikhe.

2. **Next.js project**
   TypeScript, Tailwind v4, aur Prisma ke saath. Frontend aur backend dono ek hi app mein.

3. **Database design**
   7 tables, aur access ke rules database level par.

4. **Pages aur APIs**
   9 pages aur 13 API routes — login, payment, progress, video token, admin CRUD.

5. **Video security layer**
   Signed token + stream route + watermark. Har layer alag se test ki gayi.

6. **GitHub par push**
   `github.com/preeti28dec/Youtube-Academy`

Part B — Live karna (Vercel)

7. **Vercel se GitHub repo jodna**
   Ab har `git push` par site apne aap deploy hoti hai.

8. **Database banana — Neon Postgres**
   Vercel → Storage → Create Database → Neon. Free plan, Singapore region (India ke paas), no credit card. Project se connect karte hi `DATABASE_URL` apne aap set ho jaata hai.

9. **Secrets daalna**
   `AUTH_SECRET` aur `VIDEO_TOKEN_SECRET` — Production aur Preview dono par. Inke bina app jaan-boojh kar chalta hi nahi.

10. **Database ka schema**
    Har deploy par migrations apne aap chalti hain — tables bann jaate hain.

11. **Course ka content**
    Pehli baar `/setup` page par form bhar kar syllabus aur admin account banaya. Yeh form sirf tab tak chalta hai jab tak database bilkul khaali hai.

SECTION 8

Setup ke liye jo safety net banaya gaya

Deploy karte waqt sabse aam dikkat yeh hoti hai ki configuration adhoori reh jaati hai aur site bina kuch bataye crash kar jaati hai. Isse bachne ke liye:

SITUATION	APP KYA KARTA HAI
Database ka address nahi hai	Build fail nahi hoti. Site deploy ho jaati hai aur har page par **setup screen** dikhti hai jisme likha hota hai ki kya daalna hai
Address galat hai	Setup screen uski shuruaat dikhata hai (password nahi), taaki galti turant samajh aaye
Value quotes ke saath paste ho gayi	App khud quotes hata deta hai — yeh sabse aam galti hai
Secrets nahi hain	App chalne se mana kar deta hai. Aadhe-adhoore secrets ke saath chalna asurakshit hai
Database khaali hai	Setup page par form aata hai jo shuruaati content bana deta hai

Faayda

Site kabhi bina wajah bataye band nahi hoti. Jo bhi kami hoti hai, woh screen par saaf likhi hoti hai — technical log padhne ki zaroorat nahi.

SECTION 9

Website ke pages

PAGE	KYA DIKHTA HAI	KAUN DEKH SAKTA HAI
/	Landing — benefits, curriculum, mentor, FAQ, price	Sab
/course	Poora description + syllabus	Sab
/checkout	Payment form	Sab
/login	Student login	Sab
/dashboard	Kharida hua course + progress	Enrolled student
/learn/...	Video player	Enrolled (free preview sabko)
/admin	Course aur students manage karna	Sirf admin
/guide	Client ke liye poori jaankari	Sab

SECTION 10

Admin panel — rozmarra ka kaam

Sohail ko developer ki zaroorat nahi padegi. `/admin` par login karke:

- Naya **module** (chapter) add karna
- Naya **lesson** add karna — title, duration, video
- Lesson ka naam ya duration **edit** karna
- Kisi lesson ko **free preview** banana (bina payment ke dikhega)
- Module ya lesson **delete** karna
- **Students ki list** dekhna — naam, mobile, email, kab join kiya, kitna progress
- Naam/email/mobile se student **search** karna

Video kaise jodein

Video kisi secure video host (Bunny Stream, VdoCipher, ya S3) par upload karein aur uska link admin panel mein lesson ke video field mein paste kar dein. Bas itna hi — baaki security apne aap lag jaati hai.

> **Video host kyun zaroori hai**
>
> Website ka server badi video files rakhne ke liye nahi bana. Video host se video tez chalti hai, kam kharcha hota hai, aur DRM jaisi suraksha bhi milti hai.

SECTION 11

Apne computer par kaise chalayein

Developer ke liye. Ek baar setup, phir ek command:

```
# 1. code lein
git clone https://github.com/preeti28dec/Youtube-Academy.git
cd Youtube-Academy

# 2. packages install karein
npm install

# 3. settings file banayein
cp .env.example .env

# 4. database banayein + demo data daalein
npm run setup

# 5. chalayein → http://localhost:3000
npm run dev
```

Local par database **SQLite** hai — ek file, koi server install nahi karna padta. Production par wahi code PostgreSQL par chalta hai; sirf ek line badalti hai.

Docker se bhi chal sakta hai

```
cp .env.docker.example .env.docker
npm run docker:up     # app + PostgreSQL, dono ek saath
```

SECTION 12

Abhi kya taiyaar hai, kya baaki

Taiyaar

Website live hai	Ho gaya	Vercel par
Database jud gaya	Ho gaya	Neon Postgres, Singapore
Course ka syllabus	Ho gaya	6 modules, 20 lessons
Admin account	Ho gaya	
Saare 12 features	Ho gaya	
Video security	Ho gaya	test kiya hua

Baaki

Asli videos	Baaki	Video host par upload karke admin se jodni hain
WhatsApp number	Baaki	Abhi placeholder hai
Razorpay live keys	Baaki	Abhi demo mode — asli paisa nahi katta
Sohail ki photo	Baaki	Mentor section mein placeholder
Privacy / Terms / Refund	Baaki	Payment gateway ko chahiye hote hain
Password reset flow	Baaki	Scope mein nahi tha

SECTION 13

Kharcha

Third-party prices October 2026 ke public plans par aadharit **anumaan** hain aur badalte rehte hain — quote dene se pehle aaj ke rates confirm kar lein.

MAD	ANUMAANIT	NOTE
Domain	₹800–1,500 / saal	Sirf yahi shuruaat mein zaroori hai
Website hosting	₹0 shuruaat mein	Vercel free tier
Database	₹0 shuruaat mein	Neon free tier — 1 GB
SSL (https)	₹0	Apne aap renew hota hai
Video hosting	~₹300–800 / mahina	100 students par (Bunny Stream)
Payment — UPI	Aam taur par 0%	RBI ka zero-MDR rule
Payment — Card / Net Banking	~2% + GST	₹1,499 par lagbhag ₹35
Maintenance	**Koi compulsory nahi**	Content Sohail khud admin se badal sakte hain

Shuruaat mein kul kharcha

Lagbhag ₹1,500 / saal — sirf domain. Baaki sab free tier par chal jaata hai. Students badhne par video hosting aur database paid tier par jaate hain.